
Fixing the Protein Gap

How simple food pairings turn partial carb and legume proteins into 100% usable amino acids.



Are Carb-Based Proteins Actually Usable?

The Problem

Fitness influencers say to ignore counting protein from grains and legumes. But these foods offer cheap macro support:

- 100g oats: 16g protein for ¥20
- 100g pasta: 12g protein for ¥30

The Investigation

How usable is that protein, really? I scored each grain and legume source I eat regularly against USDA amino acid data.

Optimization

How can targeted food combinations maximize the usability of lower-scoring foods without spending a fortune on extra meat?

How Protein Usability Works

The Biological Process

Dietary protein breaks down into individual amino acids to build new functional body proteins.

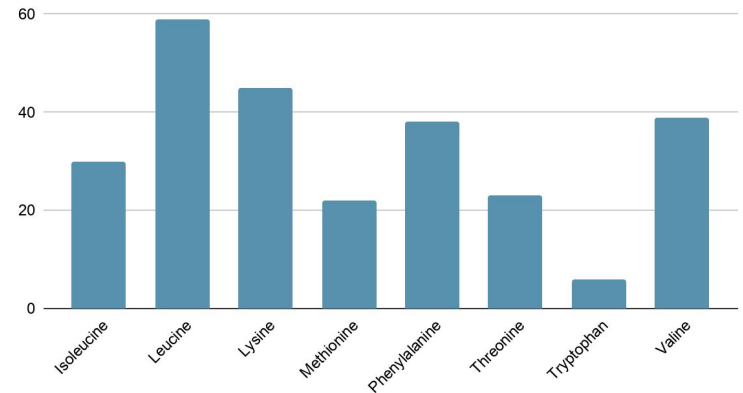
The "All-or-Nothing" Rule

Protein synthesis requires all 9 essential amino acids (EAAs) simultaneously in specific proportions.

The Limiting Factor

A shortage of even one EAA caps the total usable protein yield regardless of the total grams on the label.

Essential Amino Acids



Oats: A High-Quality Source

92.2%

of usable protein.



Limited by a 59.1mg **lysine** shortfall

Oats deliver a near-optimal essential amino acid distribution, proving that some grains function as powerful, highly efficient standalone protein sources without heavy modification.



80.3%

of usable protein.



Limited by a 60.5mg **lysine** shortfall

Rice: Same limiting amino acid

Rice has the same limiting amino acid as oats, but at a larger percentage.

Lowest protein count (6g per 100g rice) in the grains analyzed.



Pasta: Half is unused

50.8%

of usable protein.

100g of uncooked pasta carries 13.04g of protein, but only **half** can be used to build new protein. This makes the usable protein of pasta roughly comparable to 100g of rice.



Limited by a 288.8mg **lysine** shortfall



Peanut Butter also lacks lysine

68.1%

of usable protein.



Limited by a 314.8mg **lysine** shortfall

Peanut butter also lacks lysine, but at a realistic 20g serving, the true lysine shortfall is closer to 63mg.





Edamame fills the lysine gap

Optimization

50g of prepared edamame supplies 372.5mg of lysine. This more than enough to cover pasta's shortfall, along with rice, oats and peanut butter. Other super lysine-rich foods include **soybeans, lentils, and red peppers.**

New Problem

Is there a more appetizing pairing for oats and peanut butter than edamame, even if it has a little less lysine?

Greek Yogurt

50.3%

of usable protein.



Limited by a 29.7mg **tryptophan** shortfall

Greek yogurt is also lysine-rich (400mg or more per 100g). One serving fills the lysine gap for oats and peanut butter, but greek yogurt has its own shortages: **tryptophan and histidine**.

Filling the tryptophan shortage still only nets **87.1%** usable protein (19.3mg **histidine** shortfall).





Boosting greek yogurt's amino acid count

Luckily there are many delicious ways to fill the tryptophan and histidine gap:

- Mix peanut butter into the yogurt, and add crushed almonds, walnuts, or pistachios
 - Eat peanut butter on crackers or a roll alongside the yogurt
 - Oats fills yogurt's gap, and yogurt fills oats' and peanut butter's gap
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The Takeaway

- Protein completeness isn't fixed. It's fixable.
- A small pairing turns partial protein into full usable protein.
- Optimize by finding the limiting amino acid, find a food rich in it, and make a pair.
- Grain and legume proteins are affordable ways to add to daily macros

One caveat

This analysis evaluates foods in isolation rather than as part of a full daily diet. Higher-protein sources like meat supply excess amino acids that store in the body's pool throughout the day, naturally filling these grain deficits without active pairing. **However, real-time pairing provides immediate insurance for low-cost macro optimization. It guarantees maximum biological yield per meal, whether you eat meat later or not.**